

# 1 Quick Review

This week you covered the basis of every other linear algebra topic you will come across. You covered topics such as linear systems of equations, Gaussian elimination, and matrices and their operations.

A general linear system of equations looks like:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

where  $a_{ij}$  are the coefficients and  $x_i$  are the variables.

In matrix form:

$$\left[ \begin{array}{ccc|c} a_{11} & a_{12} & a_{13} & b_1 \\ a_{21} & a_{22} & a_{23} & b_2 \\ \vdots & \vdots & \vdots & \vdots \\ a_{31} & a_{32} & a_{33} & b_3 \end{array} \right]$$

## Definition 1: Matrix

A matrix is simply an array of numbers, with each number being called an *entry*.

The size of a matrix is defined as (# of rows)  $\times$  (# of columns).

If (# of rows) = (# of columns), then the matrix is a *square matrix*.

There are **three** elementary row operations:

1. Scaling: Multiplying *any* row by a scalar.
2. Interchanging: Swapping the position of two rows.
3. Replacement: Adding a constant multiple of one row to another.

These row operations are used in two elimination algorithms:

- Gaussian Elimination  $\rightarrow$  Row Echelon Form (REF)
- Gauss-Jordan Elimination  $\rightarrow$  Reduced Row Echelon Form (RREF)

## Definition 2: Inverse Matrix

The inverse matrix is defined as a matrix such that:

$$AA^{-1} = I, \text{ where } I \text{ is the identity matrix.}$$

A matrix that is invertible is also called *non-singular*.

To find the inverse of a  $2 \times 2$  matrix:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \implies A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

## 2 Problems

1. Transform the following linear systems into matrices:

(a)

$$\begin{cases} 5x_1 + 6x_2 = 33 \\ 8x_1 + 9x_2 = 34 \end{cases} \rightarrow \left[ \begin{array}{cc|c} & & \\ & & \end{array} \right]$$

(b)

$$\begin{cases} -17x_2 = 33 \\ 81x_1 + 594x_2 = 7231 \end{cases} \rightarrow \left[ \begin{array}{cc|c} & & \\ & & \end{array} \right]$$

2. Transform  $\mathbf{A}$  into a linear system of equations:

$$\mathbf{A} = \left[ \begin{array}{cc|c} 15 & 8 & 1 \\ 9 & -12 & 10 \\ -8 & 10 & 26 \end{array} \right] \rightarrow \begin{cases} x + y = \\ x + y = \\ x + y = \end{cases}$$

3. Determine the size of the coefficient matrices (make sure not to include augmented portions!).

$$\left[ \begin{array}{cc|c} a & b & c \\ d & e & f \\ g & h & i \end{array} \right] \quad \left[ \begin{array}{cccc|c} 15 & 8 & 4 & 20 & -7 \\ 9 & -12 & 9 & 24 & 23 \end{array} \right] \quad \left[ \begin{array}{ccc|c} 12 & 45 & 3 & 1 \\ 13 & -12 & 3 & 10 \\ -8 & 56 & 3 & 26 \\ 4 & 9 & 67 & 15 \\ 4 & 10 & 3 & 28 \\ -8 & 6 & 59 & 88 \end{array} \right]$$

— × —

— × —

— × —

4. Let  $\mathbf{A}$  be the matrix defined below:

$$\mathbf{A} = \begin{bmatrix} 2 & 4 \\ -3 & 7 \end{bmatrix}$$

(a) Find the inverse of  $\mathbf{A}$ .

(b) What is  $\mathbf{A}\mathbf{A}^{-1}$ ?

$$\mathbf{A}\mathbf{A}^{-1} = \begin{bmatrix} & \\ & \end{bmatrix}$$

(c) What is the name of the product of  $\mathbf{A}\mathbf{A}^{-1}$ ?

---

5. Depending on the progression of your course, you may have encountered 4 to 6 equivalent statements. Throughout the remainder of the course this list will grow **massively**.

(a) What does it mean when we say the statements are equivalent?

---



---

- (b) Do the statements apply to *every* matrix?

---

---

- (c) The statements are listed in order of implication. List the 6 equivalent statements you have discovered up to now. Use Theorem 1.6.4 in your textbook to find them:

*If  $\mathbf{A}$  is an \_\_\_\_\_ matrix, then the following are equivalent.*

i. \_\_\_\_\_

ii. \_\_\_\_\_

iii. \_\_\_\_\_

iv. \_\_\_\_\_

v. \_\_\_\_\_

vi. \_\_\_\_\_

6. **(Proof).** Let  $\mathbf{A}$  and  $\mathbf{B}$  be square matrices of the same size. If  $\mathbf{AB}$  is nonsingular, prove that  $\mathbf{A}$  and  $\mathbf{B}$  are invertible.

7. Given the following matrices **A**, **B**, and **C**:

$$\mathbf{A} = \begin{bmatrix} \frac{1}{5} & \frac{1}{5} & -\frac{2}{5} \\ \frac{1}{5} & \frac{1}{5} & \frac{1}{10} \\ \frac{1}{5} & -\frac{4}{5} & \frac{1}{10} \end{bmatrix}, \mathbf{B} = \begin{bmatrix} 2 & 7 & 8 \\ 6 & -3 & 5 \\ 10 & 1 & 2 \end{bmatrix}, \mathbf{C} = \begin{bmatrix} 4 & -2 & 1 \\ 2 & 3 & 6 \\ 5 & 7 & 1 \end{bmatrix},$$

answer the following problems.

(a) Find  $\mathbf{A}^{-1}$ .

(b) Find  $\mathbf{BC}$ .

(c) Find  $10\mathbf{A} + \mathbf{B}$ .

(d) Find  $\mathbf{CB}$ .

8. Below are two matrices  $\mathbf{A}$  and  $\mathbf{B}$ :

$$\mathbf{A} = \begin{bmatrix} 5 & 3 \\ 1 & -2 \end{bmatrix}, \mathbf{B} = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$$

(a) Verify that  $\mathbf{A}$  and  $\mathbf{B}$  commute ( $\mathbf{AB} = \mathbf{BA}$ ).

(b) Since these two matrices commute, does this mean that *all* matrices commute? If not, why?

---

---

(c) (**BONUS**). Can you find a reason why these matrices commute?